Foam rolling decreases muscle soreness but has no effects on running performance

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ABSTRACT

Muscle soreness is a concern for endurance athletes, as it may limit the ability to train and influence performance in competitions. Many athletes use foam rolling (FR) as a recovery tool, but it is unclear whether FR is effective for treating soreness in trained runners when compared to a placebo. Previous studies showing the efficacy of FR in athletes could have been confounded by a placebo effect because participants were aware that they were performing active recovery. Downhill running (DHR) has been shown to induce soreness and to impair running economy (RE).
Purpose: To study the influence of FR on soreness and running performance (RE and time-trial [TT] performance) when compared to sham tights (pants that were described to the subjects as enhancing recovery) following DHR.
Methods: Eight trained runners (ages: 31 ± 7 years; four males, four females) were recruited and completed the study. Subjects had a BMI of 23.5 ± 4.6 kg/m^2, body fat percentage of 17.9 ± 10.0, and VO_{2peak} of 57 ± 7 ml•kg^{-1}•min^{-1} (mean ± SD). Subjects performed a RE test at 65 and 75% of 5 km race speed and a 3 km TT on a track. In a repeated-measures crossover design, subjects then completed DHR followed by either a FR protocol or the use of sham tights. Two days post-DHR, subjects repeated the RE and TT tests. Two to four weeks later, subjects performed crossover visits. During RE tests, VO_2, respiratory exchange ratio (RER), and rating of perceived exertion (RPE) were recorded. Passive and active soreness were measured on a scale of 0 to 10, with 0 being no soreness and 10 being the worst soreness imaginable. Results: Both treatment conditions experienced passive and active soreness induced by DHR. Active soreness two days post-DHR was significantly lower after FR than sham tights ($p = 0.016$), whereas passive soreness approached significance ($p = 0.082$). Changes in VO_2, RER, and RPE pre- and post-DHR measures at 65 and 75% of 5 km speed did not differ significantly between treatments. Performance in the TT test also did not differ between treatment conditions. Conclusion: Foam rolling decreases leg muscle soreness in a crossover model designed to account for the placebo effect. However, FR likely does not affect running performance as measured by submaximal VO_2, RER, and RPE and by a 3 km TT.
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BACKGROUND

- Exercise-induced muscle damage (EIMD) occurs after unaccustomed exercise, particularly exercise with a large eccentric component. EIMD results in impaired muscle function, decreased range of motion (ROM), and delayed-onset muscle soreness (DOMS). These effects may impair subsequent athletic performance.
- Athletes use numerous recovery techniques, including cryotherapy, anti-inflammatory drugs, compression garments, and massage in an effort to attenuate symptoms of DOMS.
    - However, these methods are largely ineffective and may hinder both performance and the recovery process.
- Foam rolling (FR) shows some promise as a tool that promotes recovery without diminishing performance.
    - FR completed after a muscle-damaging workout alleviates soreness and prevents a decline in dynamic performance measures.
- Previous FR studies have lacked placebo controls and have not tested endurance athletes. The present randomized cross-over study uses an appropriate placebo and examines the effects of FR on well-trained distance runners.

PURPOSE

To determine whether foam rolling is an effective recovery tool to mitigate the performance-lowering effects of delayed onset muscle soreness caused by downhill running.

METHODS

Subjects
Eight well-trained distance runners (weekly mileage ≥30 for the preceding 3 months) completed the study. The average age was 30.6±6.9 years. Subjects had a mean VO_{2peak} of 57.0±7.1 ml kg^{-1} min^{-1}.

Protocol
Subjects completed a total of seven visits. At the first visit, following informed consent, body composition was determined using hydrostatic weighing. VO_{2peak} was determined using a graded treadmill protocol. Subjects were shown the FR protocol and were familiarized with DHR on the treadmill, in which the belt ran backward and subjects faced the rear of the inclined treadmill. At the second visit, 2-7 days later, running performance was assessed via a running economy (RE) test and a 3 km time trial (TT). At the third visit, 2-4 days after visit 2, subjects performed a 30-minute DHR bout. Active and passive soreness were assessed before and after the DHR using a 0-10 verbal scale. After the DHR, subjects completed the first of two recovery protocols: FR or sham recovery tights. The fourth visit occurred 48 hours after visit 3. Subjects reported their soreness, then performed a RE test and TT. The crossover took place 2-4 weeks after visit 4. At visits 5 through 7, subjects repeated visits 2 through 6, respectively, but received the alternate recovery protocol.

Assessment of Running Performance
Running performance was evaluated using a RE test and a TT. In the RE test, subjects ran a 10-minute warm-up at a self-selected speed on a level treadmill. Then, they ran two 5-minute stages, the first at 65% of their estimated 5 km race pace and the second at 75% of this pace. Stages were separated by a 2-minute standing recovery. Respiratory gases were collected throughout. Subjects rated their perceived exertion on the 6-20 Borg scale at 3 minutes into each stage. Gas data from the last 2 minutes of each stage were analyzed to determine VO_2 and the respiratory exchange ratio (RER). Subjects ran a 3 km TT at least 10 minutes after the RE test. Time trials were conducted on 200 m and 400 m tracks. Subjects received verbal encouragement each time they completed a lap.

Recovery Protocols
The FR protocol lasted for 16 minutes and was supervised by study staff. Subjects rolled the anterior, posterior, and lateral leg from hip to knee and the gluteal muscles. Subjects placed their body weight over a high-density roller at the hips and rolled their legs from proximal to distal in small motions. When reaching the knee, subjects rolled back up to the hips in one movement. They rolled each muscle group of one leg for one minute, switched legs, and repeated the protocol. In the alternate condition, subjects put on running tights. They were told that these were compression tights, which are widely advertised as promoting recovery. However, these were typical, non-compression tights. Subjects sat quietly wearing the tights for 16 minutes.

RESULTS

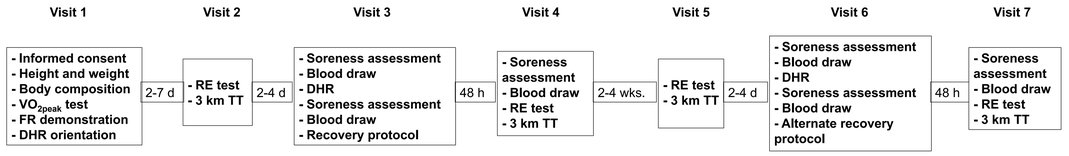


Figure 1. Study design.

Table 1. Subject characteristics.

Characteristic	Mean±SE
Males/females	4/4
Age (years)	30.6±6.9
Height (cm)	171.9±10.2
Weight (kg)	69.4±13.4
BMI (kg/m^2)	23.5±4.6
Body fat (%)	17.9±10.0
VO_{2peak} (ml kg^{-1}min^{-1})	57.0±7.1

Table 2. Percent changes in VO_2, respiratory exchange ratio, perceived exertion, and time trial time with DHR.

	VO_2 (%)		RER (%)		RPE (%)		
	65%	75%	65%	75%	65%	75%	Time Trial
Foam Rolling	-0.759	-0.740	-2.950	-1.337	-1.354	+2.239	+2.193
Tights	-2.330	+0.691	-1.330	-1.894	+7.500	+13.949	+2.504

VO_2: Oxygen consumption, ml kg^{-1} min^{-1}. RER: respiratory exchange ratio. RPE: rating of perceived exertion. 65% and 75%: Percent of subjects' 5 km race pace, used to determine the speeds of the two-stage RE test.

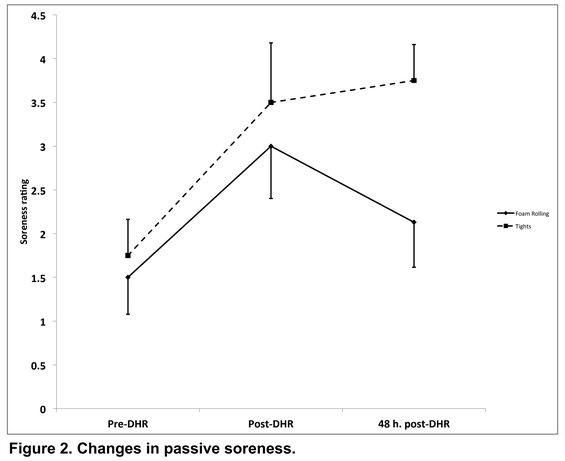


Figure 2. Changes in passive soreness.

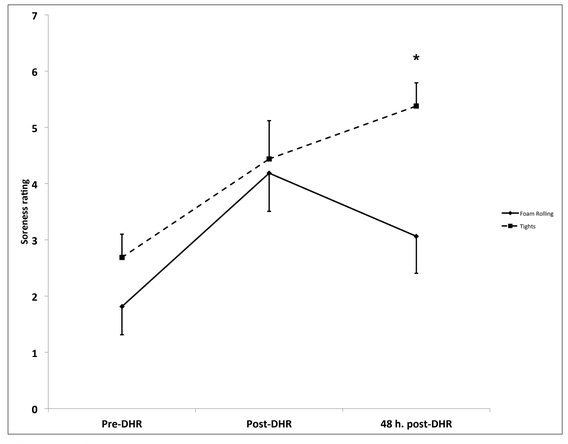


Figure 3. Changes in active soreness.
* significantly different, foam rolling vs. tights, $p < 0.05$

CONCLUSION

Downhill running (DHR) caused delayed-onset muscle soreness in both conditions, as seen by increased passive and active soreness immediately and 48 h post-DHR. Foam rolling (FR) effectively attenuated this increase in soreness compared to the sham tights. However, neither treatment affected running performance as measured by VO_2, respiratory exchange ratio, rating of perceived exertion, or 3 km time trial time. These results suggest that FR can diminish increases in soreness from exercise-induced muscle damage but does not influence running performance in well-trained runners.